

MOUSUMI DAS

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Education

University of Southern California

01/2025 - 12/2026

Master's in Electrical and Computer Engineering (Machine Learning and Data Science)

Courses - Linear Algebra for Engineering (EE 510), Probability for Electrical and Computer Engineers (EE 503)
Robot Mobility (EE 579), Robot Learning (CSCI 699), Deep Learning (EE 541)

Technical Skills

Programming & Data: Python, SQL, Git, Pandas, NumPy, Matplotlib, PyTorch, TensorFlow, Hugging Face, Scikit-learn

Relevant Skills: Robotics, Manipulation, Reinforcement Learning (RL), Policy Learning, Robot Simulation (Mujoco, robosuite), Retrieval-Augmented Generation (RAG), Deep Learning, Computer Vision, Natural Language Processing, AWS

Experience

University of Southern California

01/2026 - Present

Research Assistant to Prof. Erdem Biyik,

LIRA(Learning and Interactive Robot Autonomy) Lab

California, United States

- * Investigating teacher-aware imitation learning by modeling human demonstrations as pedagogically selected rather than reward-maximizing, and deriving a Teaching-Volume-based Behavioral Cloning framework.
- * Formulating and analyzing gradient-based update rules incorporating counterfactual teacher selection distributions to evaluate whether modeling human teaching intent improves sample efficiency over standard Behavioral Cloning.

University of Southern California

06/2025 - Present

Research Assistant to Prof. Jesse Thomason,

GLAMOR(Grounding Language in Actions, Multimodal Observations and Robots) Lab

California, United States

- * Implementing a multi-policy evaluation framework for tabletop manipulation tasks using the LIBERO benchmark, enabling selection among π_0 , OPENVLA, and keypoint-based controllers for a single task, with real-world deployment on the FRANKA PANDA arm currently in progress.
- * Developed a real-time policy selection algorithm using Implicit Q-Learning (IQL), enabling adaptive and optimal operator decision-making during wildfire evacuation dialogues.
- * Designed a context-aware RAG module that conditions response generation on the latest dialogue history following policy selection, leveraging semantic embeddings (SBERT), vector retrieval (FAISS), and instruction-tuned LLaMA-3 for adaptive generation across resident personas.
- * Benchmarked the IQL-driven policy selection model against random and fixed-policy baselines, achieving higher success rates and generalisation across diverse resident profiles.
- * Developed real-time decision integration, where dialogue outcomes trigger evacuation decisions and coordinate timely dispatch of emergency assistance once residents are persuaded or refuse to evacuate.

GlobalLogic Hitachi

03/2024 - 11/2024

Software Engineer I

Bengaluru, India

- * Implemented hidden-state probes on Vision Language Model representations to approximate hallucination evaluation metrics (factuality, consistency, modality alignment).
- * Improved inference-time efficiency of multimodal hallucination detection by replacing heavy external evaluators with lightweight probes.

California State University, Sacramento

07/2023 - 02/2024

Research Associate to Prof. Partha S Mohapatra

California, United States

- * Implemented a multi-label classification framework using cosine similarity and BART to categorize SP BSE 500 companies' CSR contributions into 13 categories.
- * Retrieved data from SEC filings via RESTful APIs, and analyzed investor risk perception by applying topic modeling (LDA) on 10-K reports using AWS-based pipelines.

Publications

- P. Padhi and M. Das, "Hand Gesture Recognition using DenseNet201-Mediapipe Hybrid Modelling," 2022 International Conference on Automation, Computing and Renewable Systems (ICACRS) [Link](#)

Projects

Lightweight On-Device AF Detection from Single-Lead ECG

09/2024 - 11/2024

- Collaborated with researchers from UC San Diego to develop and integrate a modified Lottery Ticket Hypothesis (LTH) pruning strategy to optimize both CNN and transformer architectures for on-device ECG classification
- Achieved an 80% reduction in model size while maintaining the F1 score of 0.92.

Voice and Vision Controlled Robotic Hand [Link](#)

01/2023 - 03/2023

- Developed a robotic hand interface using speech-to-text (Automatic Speech Recognition) for voice control and Mediapipe-based vision tracking for replicating human hand movements.

Automated Sortation of Packages in a Warehouse [Link](#)

10/2021 - 01/2022

- Multiple bots were monitored continuously by an overhead camera using Aruco Markers and given instructions via WiFi to the microcontroller on the bot based on their live location.